

CYBERSECURITY FUNDAMENTALS & ETHICAL HACKING BASICS

Task 1 Report

Internship: Ethical Hacking Internship

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Introduction

In today's digital world, computers, mobile devices, networks, and online services store a huge amount of sensitive information. Cybersecurity is the practice of protecting systems, networks, and data from cyber threats, unauthorized access, and attacks. As technology advances, cybercriminals continuously develop new methods to exploit vulnerabilities. Therefore, cybersecurity plays a crucial role in ensuring the confidentiality, integrity, and availability of information.

Ethical hacking is an important branch of cybersecurity that focuses on identifying and fixing security weaknesses before malicious attackers can exploit them.

What is Cybersecurity?

Cybersecurity refers to the technologies, processes, and practices designed to protect computers, networks, applications, and data from cyberattacks.

Objectives of Cybersecurity

1. Protect sensitive information.
2. Prevent unauthorized access.
3. Ensure system availability.
4. Maintain data accuracy and integrity.
5. Reduce financial and reputational losses.

Importance of Cybersecurity

- Protects personal and organizational data.
- Prevents cybercrime and fraud.
- Maintains business continuity.
- Ensures customer trust.
- Helps organizations comply with regulations and policies.

What is Ethical Hacking?

Ethical hacking is the process of legally testing computer systems, networks, and applications to identify vulnerabilities and security weaknesses.

Ethical hackers use the same tools and techniques as malicious hackers, but they operate with permission from the organization. Their primary goal is to strengthen security and protect systems from cyber threats.

Responsibilities of an Ethical Hacker

- Identify vulnerabilities.
- Perform security assessments.
- Conduct penetration testing.
- Report security flaws.
- Recommend security improvements.

Benefits of Ethical Hacking

- Prevents cyberattacks.
- Improves security posture.
- Protects sensitive data.
- Enhances customer confidence.
- Reduces business risks.

Types of Hackers

1. White Hat Hackers

White Hat Hackers are ethical hackers who work with authorization to identify and fix security vulnerabilities.

Characteristics

- Operate legally.
- Follow ethical guidelines.
- Help organizations improve security.

Example

A company hires a cybersecurity expert to test its website for vulnerabilities.

2. Black Hat Hackers

Black Hat Hackers are malicious individuals who gain unauthorized access to systems for personal, financial, or criminal purposes.

Characteristics

- Operate illegally.
- Steal data or money.
- Disrupt services.

Example

A hacker steals customer credit card information from an online store.

3. Grey Hat Hackers

Grey Hat Hackers operate between ethical and malicious hacking. They may access systems without permission but usually do not have harmful intentions.

Characteristics

- May violate rules or policies.
- Often disclose vulnerabilities after discovering them.
- Usually do not seek financial gain.

Example

A person finds a vulnerability in a website and informs the owner without prior authorization.

The CIA Triad

The CIA Triad is the foundation of information security.

1. Confidentiality

Confidentiality ensures that sensitive information is accessible only to authorized users.

Methods to Maintain Confidentiality

- Encryption
- Password protection
- Access control

- Multi-factor authentication

Example

Only authorized employees can access company payroll information.

2. Integrity

Integrity ensures that information remains accurate, complete, and unaltered.

Methods to Maintain Integrity

- Hashing
- Digital signatures
- Data validation
- Access restrictions

Example

Bank transaction records should not be modified by unauthorized users.

3. Availability

Availability ensures that systems and data are accessible whenever needed.

Methods to Maintain Availability

- Backup systems
- Disaster recovery plans
- Redundant servers
- Regular maintenance

Example

An online banking website should remain available to customers 24/7.

Common Cyber Threats

Cyber threats are activities that attempt to damage, disrupt, or gain unauthorized access to computer systems.

1. Malware

Malware is malicious software designed to harm systems or gain unauthorized access.

Types of Malware

- Virus
- Worm
- Trojan Horse
- Spyware
- Adware

Effects of Malware

- Data theft
- System damage
- Performance degradation
- Unauthorized access

Prevention

- Install antivirus software.
 - Keep systems updated.
 - Avoid suspicious downloads.
 - Scan files regularly.
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2. Phishing

Phishing is a social engineering attack that tricks users into revealing sensitive information such as passwords, banking details, or personal information.

Common Phishing Methods

- Fake emails
- Fraudulent websites
- SMS phishing (Smishing)
- Voice phishing (Vishing)

Effects

- Identity theft
- Financial losses
- Data breaches

Prevention

- Verify sender identities.

- Avoid clicking suspicious links.
 - Enable multi-factor authentication.
 - Use spam filters.
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3. Ransomware

Ransomware is malware that encrypts a victim's data and demands payment for decryption.

Famous Examples

- WannaCry
- Ryuk
- Conti

Effects

- Data loss
- Financial damage
- Business disruption

Prevention

- Maintain regular backups.
 - Update software frequently.
 - Use endpoint protection tools.
 - Train users about cybersecurity awareness.
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Ethical Hacking Process

The ethical hacking process generally follows these phases:

1. Reconnaissance

Collecting information about the target.

2. Scanning

Identifying open ports, services, and vulnerabilities.

3. Gaining Access

Testing vulnerabilities to determine security weaknesses.

4. Maintaining Access

Evaluating how attackers might persist within systems.

5. Analysis and Reporting

Documenting findings and recommending solutions.

Conclusion

Cybersecurity is essential for protecting digital assets, systems, and information from cyber threats. Ethical hacking helps organizations identify vulnerabilities before attackers can exploit them. Understanding different types of hackers, the CIA Triad, and common threats such as malware, phishing, and ransomware is fundamental to building a secure digital environment. By implementing strong security practices and promoting cybersecurity awareness, organizations and individuals can significantly reduce cyber risks.

References

1. Virtual Works Lab Ethical Hacking Internship Tutorial.
2. Cybersecurity Fundamentals Learning Materials.
3. Ethical Hacking and Information Security Concepts.